Physics 12 Winter 2026
Week 8 Discussion

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1. **Summary of Topics from Last Week**

   - *From the lectures*
     - ◇ Electromagnetic induction, Faraday's law
     - ◇ Electric motor, Electric generator
     - ◇ Direct current (DC), Alternating current (AC)
     - ◇ Power transmission, Transformer $V_1/N_1 = V_2/N_2$
     - ◇ Superconductivity
     - ◇ Energy storage (pros and cons)
       - ∗ Pumped hydroelectric storage
       - ∗ Batteries
       - ∗ Hydrogen (electrolysis, steam reforming/methane, fuel cell, etc.)
     - ◇ Electric vehicle (EV), carbon footprint, environmental impact
   - *From the textbook* (Wolfson 4th edition)
     - ◇ Electricity, Electric power usage, Power grid, AC/DC, Voltage, Current, Resistance, Electric Power, Ohm's Law, Transformer, Electric energy storage (Chapter 11.1)
     - ◇ Hydrogen (as resource, as storage), Fuel cells (Chapter 11.2)
     - ◇ Electric vehicles (Chapter 11.3)

2. **Brain-warmer/mock quiz.**

(a) **Faraday's Law.** What does a change in the magnetic flux (*area* of the loop times the *magnetic field perpendicular to the loop's area*) through a loop of wire induce in the wire loop? Select all that apply.

A. a voltage

B. a current

C. a resistance

D. an electric charge

E. heat

Answer: A and B. A change in the magnetic flux through the loop induces a voltage (potential difference) in the loop, which results in the flow of charges (current) inside the wire loop.

(b) **Hydrogen on Earth.** Which of the following statements are true about hydrogen (H_2) on Earth? Select all that apply.

A. H_2 on Earth is mostly an energy carrier, like electricity

B. H_2 on Earth is a clean, zero-emission energy resource with great potential, like wind power and solar power.

C. Most of the H_2 molecules on Earth have already escaped the atmosphere very early during Earth's history.

D. Most of the hydrogen on Earth exists in the form of water molecules.

Answer: A, C and D. Most of the H_2 on Earth has already escaped the Earth's atmosphere very early on because hydrogen molecules are very light. The hydrogen left on Earth is stored in the water molecules, which are the end product of burning hydrogen, so the hydrogen energy resource on Earth has mostly already been burned, and hydrogen can only be used for energy storage/transport.

(c) **Electric motor/generator.** What distinguishes an electric motor from an electric generator? Select all that apply.

A. An electric motor converts electric current into a magnetic field.

B. An electric generator converts a small electric potential (voltage) to a large one.

C. An electric generator converts (usually) mechanical energy into electrical energy.

D. An electric motor converts a magnetic field to an electric field.

E. An electric motor converts electrical energy into mechanical energy.

Answer: C and E. An electric generator converts (usually) mechanical energy into electrical energy, while an electric motor converts electrical energy into mechanical energy. Option B describes a transformer, not an electric generator.

(d) **Pumped hydroelectric storage.** A pumped hydroelectric storage system uses a reservoir that has a surface area of 100 km^2 and an average depth of 50 m. The pump house is 80 m below the intake, which is located at the bottom of the reservoir. Roughly how much energy can this system store? The density of water is 1000 kg/m^3, and the acceleration due to gravity is 9.8 m/s^2.

A. 4×10^9 J
B. 5×10^{13} J
C. 4×10^{15} J
D. 8×10^{13} J
E. 8×10^{10} J

Answer: C. The energy stored is

$$\begin{aligned} E &= mgh = \rho V g h = \rho A d g h \\ &= \frac{1000 \text{ kg}}{1 \text{ m}^3} \times 100 \text{ km}^2 \times \left(\frac{10^3 \text{ m}}{1 \text{ km}}\right)^2 \times 50 \text{ m} \times 9.8 \text{ m/s}^2 \times 80 \text{ m} \\ &\approx 4 \times 10^{15} \text{ J} \end{aligned} \tag{1}$$

(e) **Transformer.** Which of the following are true about a transformer? Select all that apply.

A. A transformer works with both alternating current (AC) and direct current (DC).
B. A transformer consists of two coils (primary and secondary), for which the voltage ratio between the two coils is equal to the ratio of the number of turns in each coil.
C. A step-up transformer converts a lower voltage to a higher voltage, while preserving the electric current.
D. A step-down transformer converts a higher voltage to a lower voltage, while preserving the electrical power.
E. If the primary coil has 10 turns and carries a current of 1 A, the current carried by the 100-turn secondary coil is 0.1 A.

Answer: B, D and E. A transformer only works if the current is AC, which allows for electromagnetic induction in the coils. The relationship between

the voltages, number of turns and the currents follow $V_1/V_2 = N_1/N_2 = I_2/I_1$.

3. **Hydrogen in the Atmosphere**[1]**.**

Earth's atmosphere contains 550 parts per billion of H_2 by volume.

(a) Estimate the total volume of the atmosphere, approximating it as a uniform layer 60 km thick. The radius of the Earth is 6.37×10^6 m.

(b) Use Section 11.2's figure of 12.8 MJ/m^3 to estimate the total chemical energy content of atmospheric hydrogen.

(c) If we could do the impossible and extract all that hydrogen with minimal energy expenditure, how long would it power human civilization at our current energy-consumption rate of 20 TW?

Solution:

(a) The volume of the atmosphere = area × thickness, where the Earth's surface area $= 4\pi R_E^2$, and Earth's radius $= R_E = 6.37 \times 10^6$ m:

$$V = (\text{area}) \times (\text{thickness}) = 4\pi(6.37 \times 10^6 \text{ m})^2 \times 60 \times 10^3 \text{ m} = 3.1 \times 10^{19} \text{ m}^3 \quad (2)$$

(b) The chemical energy content of atmospheric hydrogen is the energy density of hydrogen, 12.8×10^6 J/m^3, times the total volume of the atmosphere, times the 550-parts-per-billion fraction of the atmosphere that's H_2:

$$\begin{aligned} \text{energy}_{H_2} &= \frac{\text{energy}}{\text{volume}} \times \text{volume}_{H_2} \\ &= 12.8 \times 10^6 \text{ J/m}^3 \times (3.1 \times 10^{19} \text{ m}^3) \times \frac{550}{10^9} = 2.1 \times 10^{20} \text{ J} \end{aligned} \quad (3)$$

(c) Power = energy per time, so time = energy per power. Humans use about 20 TW of power, so the time to consume the hydrogen's energy would be

$$\frac{2.1 \times 10^{20} \text{ J}}{20 \times 10^{12} \text{ J/s}} = 1.05 \times 10^7 \text{ s} = 0.33 \text{ years} = 4 \text{ months} \quad (4)$$

[1] Wolfson (4th edition) Chapter 11 Exercise 7